



Flying car service

A startup flying car service, Kitty Hawk, backed by Google co-founder Larry Page has begun offering test flights. <https://dgit.in/jul18-24-1>



S-Ray under development

Samsung's S-Ray speaker beams sound only the user can hear, and could possibly replace headphones. <https://dgit.in/jul18-24-2>



Indian Regional Navigation Satellite System or, as it is operationally referred to as, the NavIC. We have reached the final stages of the program, to the point where NavIC promises greater locational accuracy than GPS. The GPS constellation provides an accuracy of 20 meters, while the NavIC constellation provides an accuracy of 5 meters. The journey to this point officially started more than a decade ago.

THE ANNOUNCEMENT

Back in 2007, about a decade after the GPS request was denied by the US, the Government of India announced that ISRO will be building and launching a constellation of 7 satellites as a part of India's own indigenous navigation network, joining the ranks of countries such as USA, Japan, Russia and China. At that point of time, the project had been set a deadline of 2012 to become operational, with the first launch planned on the PSLV in 2010. Initially, the project was called the Indian Regional Navigation Satellite System (IRNSS). The goal of 7 satellites was set because India was looking to cover the country's own lands as well as a surrounding distance of 1500 kms from the border. A similar global network would require around 24 satellites. By 2010, the various subsystems of the satellites were being manufactured by ISRO at various locations and the first launch was scheduled for late 2011 on the PSLV, and then one launch every six months. By 2014, with four satellites active (the minimum required) the NavIC system was scheduled to be operational.

THE LAUNCHES BEGIN

As it would happen, the actual launches began behind schedule for various reasons. On May 28, 2013, the ISRO Navigation Centre was inaugurated at the Indian Deep Space Network complex at Byalalu, about 40 kilometers from Bangalore, India. The Navigation Centre was supposed to be a crucial point of operation for providing the time reference, generation of naviga-

The NavIC story

In its decade long journey to near-completion, the NavIC project has seen a lot of highs – and even some lows

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very country stands to benefit in multiple ways from technological advancements. At times, more than the benefits, it is an external compulsion that forces the progression of

technology. During the Kargil War between India and Pakistan, when India asked the United States of America to share global positioning system (GPS) navigation data for the region, the request was denied citing the conflict. This particular action sped up India's march towards developing and launching its own space based navigation system, the